

HealthConnect Clinic Experience Lab — Week 7 Analytics Testing & Refinement Report

Section 1: Analytical Testing & KPI Validation

Week 7 marks the transition from integration to systematic testing, visual refinement, and end-to-end validation of the HealthConnect Decision Support System (HealthConnect_Experience_Lab_Dashboard_v2.pbix).

1.1 KPI Calculation & Raw Data Cross-Verification

KPI Card Metric	Dashboard Visual Output	Raw CSV Query Result	Variance	Status	Underlying Calculation Logic
Total Appointments	5K	5,000 records	0.00%	PASS	COUNTROWS(Appointment_Data)
Overall No-Show Rate	48.46%	48.46%	0.00%	PASS	DIVIDE(Total_NoShows, Total_Appointments)
Total No Shows	2K	2,423 appointments	0.00%	PASS	CALCULATE(COUNTROWS(), Status="No-Show")
Total Lost Revenue USD	\$121K	\$121,150.00	0.00%	PASS	SUMX(NoShows * 50)

1.2 Interactive Filter & Edge-Case Stress Testing

- **Lead Time × Past Attendance Interaction Test:** Selecting **30+ Days** lead time combined with **1+ Past No Shows** correctly isolates the critical **67.88%** attrition peak, matching raw matrix sub-segment calculations.

- **Filter Combination Stability Test:** Toggling between Follow-up appointment types and No reminder channels dynamically adjusts the total lost revenue figure without altering the baseline row-level architecture.

Section 2: Dashboard Usability & Visual Refinement

Visual Component	Testing Issue Identified	Action Taken & Refinement Applied	Retest Outcome
KPI Header Cards	Display units (2K, \$121K) rounded off exact operational figures.	Changed visual display units to None to render 2,423 and \$121,150.	Exact figures are immediately visible without loss of precision.
2D Risk Matrix	Matrix rows sorted alphabetically by lead time.	Applied custom DAX sort-by-column index (Same Day → 1-7d → 8-14d → 15-30d → 30+d).	Rows display in chronological operational sequence.
Appointment Type Chart	X-axis labels truncated (Specialist Consult...).	Expanded Y-axis label category width in formatting panel.	Full text (Specialist Consultation, General Consultation) visible.
Donut Visual	Slices displayed low contrast between Attended and Cancelled.	Adjusted color palette to high-contrast accessible healthcare hex codes.	Clear visual distinction across all three outcome categories.

Section 3: Cross-Track Testing & Validation (Data Analytics → Data Science)

Testing Cycle Workflow: Test Finding → Action Taken → Retest Result → Validated Improvement

1. **Test Finding:** Data Science models required lead time categorized into non-linear behavioral risk bins rather than continuous days to optimize classification accuracy.
2. **Action Taken:** Validated the lead time distribution in Power BI and structured the exported feature matrix (ds_da_feature_matrix_v2.csv) into explicit intervals (0–6d, 7–13d, 14–27d, 28+d).

3. **Retest Result:** Verified that DAX measures and Data Science model pipeline ingestion scripts produced identical patient bucket totals without unassigned missing records.
4. **Validated Improvement:** Established a unified risk hierarchy across both analytical dashboards and machine learning prediction inputs.

Section 4: Testing & Validation Log

Component Tested	Testing Objective	Expected Result	Actual Result	Status	Action Taken	Retest Result
Risk Matrix Visual	Validate 30+ day attrition peak	67.88% peak displayed	67.88% displayed	PASS	Fixed row sort order	Chronological lead time display
Revenue Loss Card	Validate total financial impact	\$121,150.00 calculated	\$121K rounded display	PASS	Formatted card to exact USD	Displayed \$121,150 clearly
Reminder Channel Visual	Identify top no-show channel	No reminder at top	51.39% (No reminder)	PASS	Adjusted data label padding	Labels fully readable
Slicer Cross-Filtering	Stress-test multi-select filters	Stable metric calculation	No calculation errors	PASS	Locked visual interaction rules	Verified smooth drill-downs

Section 5: Limitations & Week 8 Recommendations

- **Current Limitations:** The dataset lacks explicit timestamped appointment cancellation reasons, limiting qualitative analysis on why the 30+ day lead-time segment exhibits high attrition.
- **Week 8 Final Integration Recommendations:**
 1. Integrate the Data Science track's predicted no-show probabilities directly into the Power BI model as a real-time risk score column.
 2. Embed automated alert thresholds for administrative staff when booking appointments with lead times exceeding 30 days.